Team Name - Rockers

Hindi Fluency and Semantic Organization: Advanced Statistical Analysis

BRSM Spring 2026 | IIIT Hyderabad | Verbal Fluency Task + Spatial Arrangement Method

BRSM PROJECT POSTER PRESENTATION SP 2026

ABSTRACT

This study investigates lexical retrieval and semantic organization in Hindi speakers using the Verbal Fluency Task (VFT) and Spatial Arrangement Method (SpAM). We analyzed data from 35 participants across four semantic categories (Animals, Foods, Colours, Body Parts), collecting 1,044 word responses and 1,040 spatial placements.

Using ANOVA, Multiple Regression, Generalized Linear Models, and Bayesian statistics, we found:
- Word production differs across categories ($p = .009$)
- Hindi proficiency predicts retrieval speed but not word count (ANCOVA $p = .007$; Bayesian $BF_{10} = 0.091$)
- Spatial distance predicts IRT after controlling for category ($\beta = 0.202$, $p = .044$)
- Serial position effect: within-cluster acceleration ($\beta = -0.017$, $p = .002$)

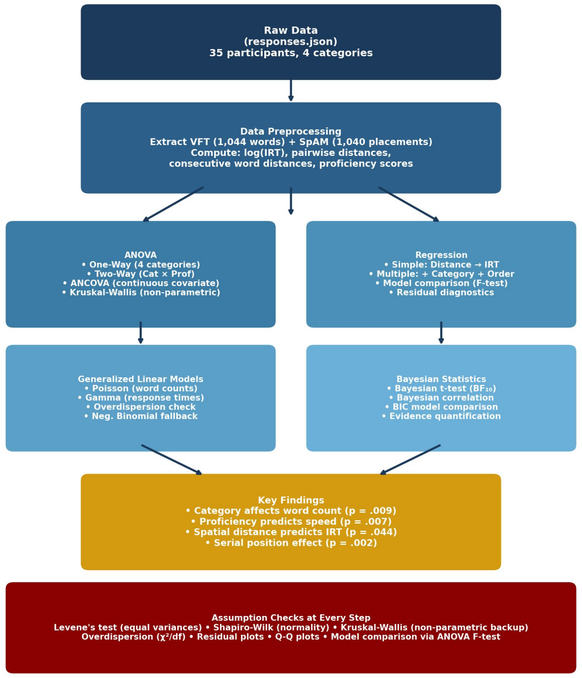
These findings highlight how choosing the right statistical methods can reveal effects that simpler analyses miss.

METHODOLOGY

Data: N = 35 participants; 4 categories; 1,044 word responses; 1,040 SpAM placements. Tasks administered online via jsPsych.

Preprocessing: Extracted VFT (words + IRT) and SpAM (x,y coordinates) from JSON. Computed pairwise distances, consecutive word distances, log-transformed IRT, and Hindi proficiency composites (average of reading & writing ratings).

Significance level: $\alpha = 0.05$ for all tests. Tools: Python (scipy, statsmodels, pingouin).

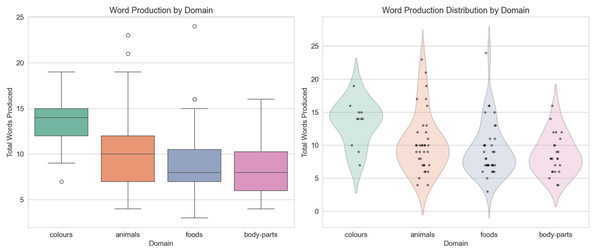


RESULTS

1. ANOVA — Word Production Across Categories

One-Way ANOVA: $F(3,101) = 4.05$, $p = .009$, $\eta^2 = .107$ (medium effect)
- Colours (M=13.4) > Body Parts (M=8.6): $p = .008$, Hedges' $g = 1.47$
- Colours (M=13.4) > Foods (M=9.3): $p = .022$, Hedges' $g = 1.01$
- Welch's ANOVA: $F(3,39.6) = 5.44$, $p = .003$
- Kruskal-Wallis (non-parametric): $H = 13.10$, $p = .004$

Colours is a smaller, bounded category allowing near-exhaustive retrieval.



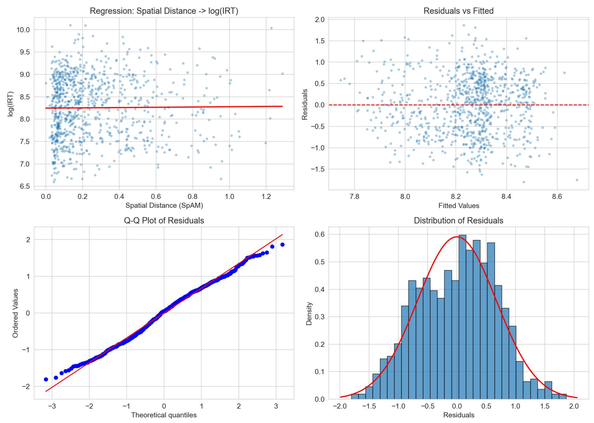
2. ANCOVA — Proficiency Effect Recovered

Report 1 found proficiency → IRT was NOT significant ($p = .060$).
ANCOVA keeps proficiency as a continuous variable (not dichotomized):
- Category: $F(3,100) = 3.28$, $p = .024$, $\eta^2 = .090$
- **Hindi Proficiency: $F(1,100) = 7.60$, $p = .007$, $\eta^2 = .071$**

Higher proficiency → faster retrieval. Category-level variance was hiding this effect in Report 1.

3. Regression — Spatial Distance Predicts IRT

Simple regression (no controls): $\beta = 0.031$, $p = .750$ (NOT significant)
After controlling for category:
$\beta = 0.202$, $p = .044$ (SIGNIFICANT)
Adding serial position: $\beta_dist = 0.225$, $p = .025$; $\beta_order = -0.017$, $p = .002$
Model comparison: $F(4) = 12.35$, $p < .001$
→ Category was a confounding variable masking the real effect.

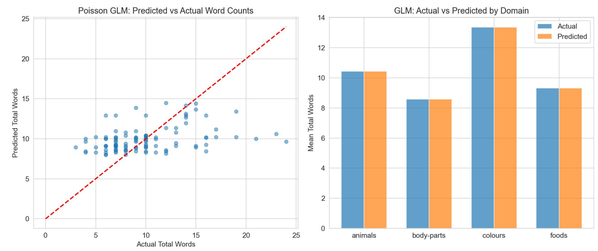


4. Generalized Linear Models

Poisson GLM (word counts are integers, not continuous):
- Colours: 28% more words than Animals ($p < .001$)
- Body Parts: 19% fewer words than Animals ($p < .001$)
- Mild overdispersion ($\varphi = 1.61$); Neg. Binomial gave same results.

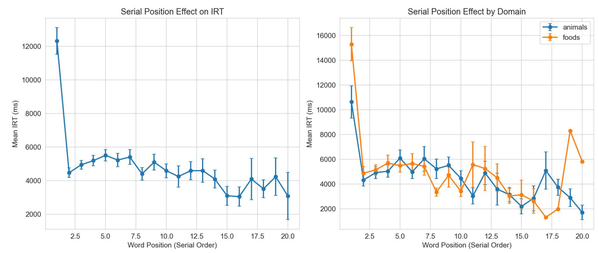
Gamma GLM (IRT is always positive & right-skewed):
- **Hindi Proficiency: $\beta = -0.091$, $p = .025$**
  Each 1-point increase → 8.7% faster retrieval
- Total Words: $\beta = -0.027$, $p = .045$ (more fluent = faster)
- Colours domain: 29% faster than Animals ($p = .007$)

GLMs found effects OLS missed by correctly modeling data distribution.



5. Serial Position Effect

$\beta = -0.017$, $t = -3.18$, $p = .002$
Each position → ~1.7% faster retrieval.
Pattern: startup → within-cluster acceleration → slowdown at ~position 12.



6. Bayesian Analysis

- Bayesian t-test (IRT by proficiency group): $BF_{10} = 1.02$
  → Inconclusive with dichotomized groups.
- Bayesian correlation (spatial dist vs IRT): $BF_{10} = 0.064$
  → Strong evidence for NO simple effect.
- BIC model comparison (proficiency → word count): $BF_{10} = 0.091$
  → Strong evidence proficiency does NOT predict quantity.

Takeaway: Proficiency affects SPEED but NOT QUANTITY.

CONCLUSIONS

1. Category matters for word production.
Colours produces significantly more words than Body Parts ($p=.008$) and Foods ($p=.022$). Bounded categories allow near-exhaustive retrieval.

2. Proficiency predicts speed, not quantity.
ANCOVA ($p=.007$) and Gamma GLM ($p=.025$) recovered a proficiency→speed effect that Report 1 missed. Bayesian analysis ($BF_{10}=0.091$) confirms proficiency does NOT predict word count.

3. Spatial distance predicts IRT after controls.
Simple correlation ($p=.750$) masked a suppression effect. Controlling for category reveals the true relationship ($\beta=0.202$, $p=.044$).

4. Serial position drives retrieval dynamics.
Within-cluster acceleration ($\beta=-0.017$, $p=.002$): ~1.7% speedup per position.

5. Statistical model choice matters.
Poisson/Gamma GLMs found effects that ordinary regression missed by correctly modeling count data and skewed response times.

FUTURE WORK

- Mixed-effects models with random intercepts for participants to handle nested data structure.
- Automatic cluster detection using DBSCAN on SpAM coordinates.
- Apply Log transformation to IRT data, to ensure normality assumption of linear regression.
- Cross-linguistic comparison (Hindi vs English) within the same bilingual participants.

TEAM CONTRIBUTIONS

Deban Kumar Shahi (2024201048)
ANOVA analyses, regression modeling, assumption testing, model diagnostics.

Md Fraz Uddin (2024201064)
Python pipeline, figures, GLM & Bayesian analyses, report & poster.

Suraj Lalwani (2024201071)
Data extraction from JSON, SpAM metrics, preprocessing pipelines.

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